

EMPOWERING INTERDISCIPLINARY EXPERTISE: EVALUATING THE IMPACT OF CHALLENGE-BASED LEARNING ON AI AND SUSTAINABILITY LITERACY IN HIGHER EDUCATION

B. Zönnchen¹, C. Böhm¹, H. von Schwichow², G. Socher¹, S. Wurster²

¹HM Hochschule München University of Applied Sciences (GERMANY)

²Technical University of Munich (GERMANY)

Abstract

In the context of rapidly evolving discussions on sustainability and artificial intelligence (AI), understanding the synergies and tensions between these critical areas is increasingly vital. The growing effort towards AI for sustainability must be complemented by the sustainability of developing and using AI systems. To master the associated challenges and to be able to leverage its potential, we will need experts who possess specific skills to sufficiently understand, analyze, and ultimately shape complex systems soon. Addressing this need requires interdisciplinary teaching and learning offerings that enable students to work independently as well as collectively, network knowledge, and come to informed decisions and solutions to complex problems. Consequently, a new teaching format has been implemented: sustAIability – advanced topics on sustainability and AI. This initiative, a collaborative effort by Technical University of Munich and Hochschule München University of Applied Sciences, funded by Bavarian Institute for Digital Transformation (bidt) for four years, utilizes challenge-based learning to immerse students in real-world problems sourced from industry, academia, and the public sector. The seminar features four intensive sessions exploring the sustainability of AI and AI for sustainability from economic, ecological, and social perspectives, highlighted through expert examples. Students then engage with real-world challenges presented by partners from research, public administration, industry, and civil society, focusing on sustainability and AI. To guide students through the process, various methods including design thinking are incorporated into the seminar. The pressing question is whether, in such a condensed time frame, students can cultivate and enhance their efficacy in sustainability and AI by addressing real-world-challenges?

This paper examines the effectiveness of this methodology in terms of the students' self-efficacy regarding the combination of sustainability and AI, as well as the learners' willingness, abilities, and behavior to contribute to the Sustainable Development Goals (SDGs). Therefore, we conducted a preliminary evaluation of the second cohort during the winter semester of 2023/24, where 20 out of 32 participants responded after completing the seminar. The responses indicate that, through challenge-based learning in our seminar, students' literacy and self-efficacy in sustainability and AI have improved, regardless of their background or prior knowledge. We conducted a second survey, which is part of a pre- and post-evaluation, aimed at measuring learners' willingness, abilities, and behavior to contribute to the SDGs. 30 out of 48 participants voluntarily took part in the survey. It shows that the designed questionnaire has good potential to improve educators' understanding of Sustainable AI-related abilities, attitudes, and actions of cohorts of students within higher education.

Keywords: challenge-based learning, sustainability, artificial intelligence, design thinking.

1 INTRODUCTION

Universities are currently experiencing significant disturbances from their external environment. Notable among these are demographic shifts, the rapid acceleration of digital transformation, and changing expectations from the public, industry, and government. One of the most significant causes of these disturbances is the demanded (and needed) transformation toward a sustainable society. Consequently, traditional academic practices and teaching habits are being questioned. Understood as innovative ecosystems [1], universities must adapt their organizations to continue to thrive in their environment. They must also take on social responsibility [2] and thus support sustainable development [3] by promoting social [1] as well as technical innovations that promote sustainability but are also sustainable by design.

Students are anticipated to manage complexity and continual changes in the workplace, concurrently tackling issues related to sustainability [4]. Knowledge, skills and dispositions of higher education students need to be re-oriented towards sustainability. A desired outcome of education for sustainable

development (ESD) are action competences, which can be defined as the relevant knowledge, willingness, behavior and self-efficacy for contributing to solving controversial problems [5]. Furthermore, to prepare future innovative entrepreneurs, leaders, and engineers for existing challenges and the rapid development of societal needs, they require knowledge and skills that include adaptability, flexibility, critical thinking, interdisciplinary collaboration, communication, complex problem solving, and systems thinking [6]. To develop these competencies, sustainability must be considered a continuous process. Educators should inspire their students to recognize their roles in sustainable development and engage in multidisciplinary collaboration to exchange knowledge. Consequently, the required competencies can no longer be imparted through traditional teacher-centered teaching units [7]. In other words, educational innovation is needed.

One of the teaching methods often employed in higher education for sustainable development are the so-called active learning methods [8]. These include problem-based (PBL), project-based (PjBL), and challenge-based learning (CBL), with CBL having evolved from PBL [7]. All these methods can enhance technical as well as cross-disciplinary skills such as teamwork [9], communication [10], critical thinking [11], and self-directed learning [12]; however, they require a paradigm shift that poses significant challenges for both students and educators. Under these student-centered teaching methods, knowledge is no longer transferred through passive participation but actively constructed by students themselves. Educators create the necessary conditions and act as facilitators and co-constructors of, and for learning [13].

Along these lines, challenge-based learning (CBL) is particularly promoted by the European Consortium of Innovative Universities (ECIU) and, according to Sukacké et al. [7], is particularly suited among the methods for education aimed at sustainable development. However, it is also one of the newer thus least researched methods. Compared to PBL and PjBL, there is a clear lack of research regarding the effectiveness of CBL. It is also important to note that active learning methods are under criticism regarding their implementation, practice, and research [7]. It is difficult to understand the process and to measure the effectiveness of active learning methods beyond the students' performance and products of learning. Furthermore, especially novice problem solvers are motivated by finding a solution rather than gaining a clear understanding of the task [14]. Thus, there is a tendency to minimize effort by skipping the developing a deep understanding and producing surface solutions [15]. Since all three methods share a common basis, some research results are interchangeable. However, there are differences worth investigating. CBL distinguishes itself by confronting learners with real-world challenges that require innovative, creative, and multidisciplinary interventions thus collaboration with various external stakeholders and experts is necessary. In their literature review, Sukacké et al. [7] highlight the importance of the students' freedom to choose and take ownership of real, unresolved challenges but also emphasize CBL's strong dependency with prior knowledge, intrinsic motivation of students and educators, support in teamwork, and a suitable learning environment created by educators and external training partners. CBL seems to work only effectively under the right conditions which necessitates to overcome possible resistances from students, educators as well as other professionals and institutions.

Against this background, our study aims to address the aforementioned research gap as well as the implementation of CBL as an innovative pedagogy in the context of sustainability and artificial intelligence (AI). As a basis, we establish and investigate the educational effectiveness of the so-called Digitalization Lab sustAIability - Advanced Topics on Sustainability and AI - that was established as one of 13 labs by the Bavarian Research Institute for Digital Transformation (bidt). With sustAIability, the university responds to yet another source of disturbance: the rapid development and outstanding successes in the field of deep learning and the inevitably associated potentials and risks for sustainable development [16,17]. We concur with Crawford and Calo [18] that AI signifies both a cultural and a technical shift. Consequently, the need to connect technological progress with sustainability within an interdisciplinary and culturally varied setting is increasingly imperative. The sustAIability course focuses on Sustainable AI [19] by, on the one hand, promoting the development of innovative AI solutions and, on the other, exploring the sustainability of AI systems critically.

Our study sought to answer the following two research questions:

- Q1. How to implement CBL to strengthen education for Sustainable AI?
- Q2. How effective is CBL in the context of Sustainable AI?

2 METHODOLOGY

2.1 Course Structure

The course is designed for master's and advanced bachelor's students from both technical and non-technical disciplines, explicitly adopting an interdisciplinary approach. It is structured into three phases: a self-learning phase, an input phase, and a workshop week. During the self-learning phase, students access relevant academic literature, as well as current newspaper articles, podcasts, and specially produced instructional videos. Following the flipped classroom model [20], students independently gain an overview of the intersection between sustainability and AI, deepening their knowledge through initial assignments.

In the subsequent four-week input phase, students attend half-day course sessions during which they receive specific insights into ongoing debates concerning AI and sustainability. These sessions consist of coach-led inputs, expert lectures with guests from the university ecosystems and beyond, and interactive exercises in which students discuss acquired knowledge and present it to each other following the learning by teaching principle [21]. Following the input phase, students are introduced to real-world challenges at the intersection of AI and sustainability, presented by partner organizations from research, public administration, industry, and the public sector. In this context, students interact with representatives from these organizations, have the opportunity to ask questions, select a challenge, and subsequently, are assigned to a team.

Challenge selection occurs prior to the course and is guided by coaches. Organizations proposing challenges can apply to participate in the course and define their respective challenges using a predefined questionnaire. Feedback sessions with coaches refine and prepare the challenges.

In total, six different challenges were offered in the summer semester of 2023 (including challenges contributed by SAP, OECD, and the start-up Imagify). In the winter semester of 2023/24, students could choose challenges from five different organizations, some of which were continued from the previous semester. Challenges addressed varied issues at the intersection of AI and sustainability, such as improving a machine learning model to identify ocean debris using satellite data or proposing enhancements to the social dimension of sustainability in the European AI Act. Due to the interdisciplinary nature of the program, both technical and non-technical solutions are encouraged as outcomes of student work, depending on the challenge.

The presentation and selection of challenges is followed by a reflection week, during which students review and organize the knowledge gained in the self-learning and input phases and prepare for work on the challenges. Additionally, students use this week to familiarize themselves with their team members, conduct research, exchange initial ideas, and schedule an initial meeting with challenge partners.

The intensive workshop during which students work for five full days on innovative solutions for their respective challenges, is the highlight of the course. The format is designed like a hackathon or ideation sprint, aiming to support students in developing high-quality solutions within a limited timeframe. Students are provided with a toolkit of agile methods, such as Design Thinking, and guided through the process with diverse exercises on idea generation, reflection, and prototyping. Regular feedback sessions with coaches, involved professors, technical experts, and challenge partners support the teams throughout.

The workshop week concludes with a public pitch night, where students present their results to challenge partners, peers, and other interested attendees. The perpetuation of the student works on the challenges is a key factor of this course and was realized in some instances: several students had the opportunity to pursue their ideas through work-study activities, as for example the TUM Venture Labs challenge. Furthermore, some of the student ideas have been adopted by the challenge partners and partially implemented in practice. For instance, suggestions for additions to the European AI Act were forwarded to the European Parliament through the Center for Ethical Machine Decision Making for further discussion.

2.2 Participants

In the first iteration in summer 2023 a total number of 43 students participated in the course, of which 29 were enrolled at Technical University of Munich (TUM), 10 students at Hochschule München University of Applied Sciences (HM), and 4 students at Ludwig-Maximilians University Munich (LMU). The sample of the first cohort consisted of 10 bachelor's and 33 master's students with different

academic backgrounds. As this course is offered at all different universities as an elective module, student backgrounds vary from Social Sciences, such as the master's programs Politics & Technology (TUM) or Master Architecture (TUM), Human-Computer-Interaction (LMU) or Computer Sciences (HM).

The cohort of the second iteration consisted of 27 Master students and 5 Bachelor students, 19 students from TUM and 13 students from HM. Again, their study programs varied from technical degrees to social sciences with Master Politics & Technologies (TUM) to Bachelor Informatics and Design (HM).

On April 26th, 2024, the third cohort started with 14 Bachelor students and 34 Master students, of which 26 are enrolled at HM 20 at TUM and two at LMU.

The most frequent majors studied by participants of the first two iterations were Politics & Technologies (in total 28 students) and Computer Sciences (in total 10 students) during both semesters, as shown in the following graph.

2.3 Measurement Instruments

ESD and change programs, such as sustAInability, need measurement instruments to monitor learning outcomes. Developing suitable instruments to measure sustainability competencies is an ongoing research effort. We decided to create our own instruments: one to measure the perceived self-efficacy after finishing the course, and another to measure action competencies for sustainability of our target group.

2.3.1 Measuring the students' self-efficacy

We conducted a questionnaire to measure the students' self-efficacy regarding the combination of sustainability and AI of the second cohort in the Winter Semester 2023/24. We wanted to know how they perceived their learning progress. The questionnaire consisted of the questions in Table 1.

Table 1. Mapping between the question abbreviations and their content.

Abbr.	Question
Q1	What was your initial motivation to participate in this course? Please explain.
Q2	Participating in the course has enhanced my knowledge and skills about sustainability of AI.
Q3	Participating in the course has enhanced my knowledge and skills about AI for sustainability.
Q4	I want to continue developing my knowledge and skills about AI applications for the Sustainable Development Goals (SGDs) after this course (additional classes, project continuation after the course, career orientation, etc.)
Q5	I feel capable of applying AI concepts and techniques to impact the Sustainable Development Goals (SGDs).
Q6	What would have been your answer to the previous question before the course?
Q7	I benefited from the interdisciplinary nature of the project workshop course.
Q8	Please explain your previous answer.
Q9	The challenge-based format of the course specifically helped me to develop my knowledge and skills about sustainability and AI.
Q10	Please explain your previous answer.
Q11	I am able to assess how an AI contributes and hinders the Sustainability Development Goals (SGDs).

The questionnaire utilizes Likert scales, ranging from -2 (strongly disagree) to 2 (strongly agree). Additionally, it incorporated three open-ended questions, identified as Q1, Q8, and Q10. Questions can only be answered after completing the course.

2.3.2 Measuring the students' disposition, abilities, and behaviors

In addition, we provided another questionnaire at the beginning of the course in the summer semester 2024. Instead of evaluating the students' gained self-efficacy, it is designed to understand or profile the learners' willingness, abilities, and behaviour to contribute to the Sustainable Development Goals (SDGs).

Table 2. Mapping between the question abbreviations and their content.

Abbr.	Question
GE01	What was your initial motivation to participate in this course? Please explain.
GE02	I am interested primarily in artificial intelligence.
GE03	I am interested primarily in sustainability.
GE04	I think artificial intelligence is beneficial for achieving a more sustainable future.
GE05	I think artificial intelligence is an obstacle to achieving a more sustainable future.
GE06	Please name the most important aspect of artificial intelligence that hinders the realization of the Sustainable Development Goals (SDGs).
GE07	Please name potential opportunities for artificial intelligence to realize the Sustainable Development Goals (SDGs).
GE08	Put the following list of activities in order of their environmental impact from largest to smallest: (1) Flying in a commercial airplane from Washington D.C. to China, (2) Producing one fast food quarter-pound hamburger, (3) Producing one fast food chicken sandwich, (4) Keeping a cell phone charger plugged into an electrical outlet for 12 hours
WI01	I am willing to develop my knowledge and skills about AI applications for the Sustainable Development Goals (SDGs).
WI02	I am willing to listen to other people's views.
WI03	I am willing to participate in an initiative to make my community sustainable.
WI04	I am willing to change my lifestyle to embrace sustainable living.
AB01	I am able to apply AI concepts and techniques to impact the Sustainable Development Goals (SDGs).
AB02	I am able to assess how AI contributes to and hinders the Sustainable Development Goals (SDGs).
AB03	I am able to identify the causes of my community's unsustainable development.
AB04	I am able to assess the impact of my personal lifestyle on the environment.
AB05	I am able to assess the impact of my government's development policies on my country's future sustainable development.
AB06	I am able to develop business ideas involving artificial intelligence to positively impact the realization of the Sustainable Development Goals
AB07	I am able to develop a strategy to improve people's quality of life.
BE01	I advocate for my local community's sustainability.
BE02	I encourage people to engage in the community's sustainability activities.
BE03	I advocate for environmental justice.
BE04	I avoid using fossil fuels for traveling.
BE05	I try to reduce my ecological footprint.

Like the Sustainability Consciousness Questionnaire (SCQ-L, SCQ-S) [22] and the Self-Perceived Action Competence for Sustainability Questionnaire (SPACE-Q) [23], our aim was to measure action competence for sustainability [24]. However, we decided to create our own questionnaire which is

inspired by the DAB (Dispositions, Abilities, Behaviours) framework [25]. It consists of qualitative and quantitative instruments to understand the five clusters of sustainability competencies: (1) learning to know, (2) learning to do, (3) learning to be, (4) learning to live together, and (5) learning to transform oneself and society. The reason behind this decision is that we wanted to include aspects of Sustainable AI into the items, especially within the ability category (AB), and we wanted to use items that are more suitable for master or advanced bachelor students who are already interested in the topic.

To maximize participants' response rates, we streamlined the questionnaire to a total of 24 items, distributed across four categories. The items are depicted in Table 2. The first segment, consisting of eight questions (GE01-GE08), explores the students' motivation and prior knowledge. Subsequent sections probe learners' willingness to act (four questions, WI01-WI04), abilities in Sustainable AI (seven questions, AB01-AB07), and behaviours that support sustainability (five questions, BE01-BE05). Each category is structured on a Likert scale ranging from -2 (strongly disagree) to 2 (strongly agree), allowing for nuanced analysis of each competency dimension. To compare the learners' progress, we are planning to additionally conduct a similar questionnaire after the course.

3 RESULTS

In this section, we detail the outcomes of our surveys and the adjustments we made based on feedback from the coaches. We begin by presenting the evaluation results from a self-assessment questionnaire completed by 20 out of 32 participants. Then we present the results of our second instrument which is inspired by the Disposition, Abilities, and Behaviours (DAB) framework. Following this, we explore the changes implemented for the current cohort.

3.1 Assessing learners' perceived self-efficacy

Data illustrated in Fig. 1 reveal that (N=20) participants reported perceived enhancements in their understanding of AI's sustainability (Q2) and the application of AI to sustainability efforts (Q3). A predominant number of participants expressed a sustained interest in the domain (Q4). Moreover, there was a noted increase in the respondents' ability to apply their knowledge effectively (Q5-Q6) and to assess the potential challenges and benefits associated with the Sustainable Development Goals (SDGs) (Q11). The course's interdisciplinary and challenge-based learning framework received positive feedback from the respondents (Q7-Q10). The underlying motivation for student participation in the course (Q1) reflects the significant role they attribute to AI in their future professional endeavors. Specifically, four students primarily highlighted their interest in sustainability, whereas seven pointed to AI as the primary motivator for enrolling in the course, and eight acknowledged both sustainability and AI as key factors.

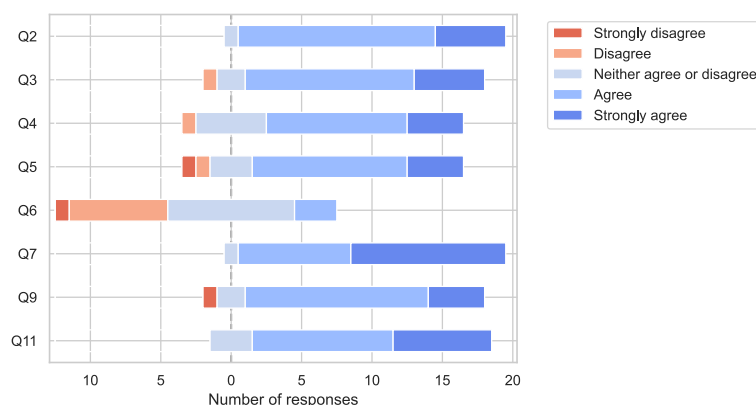


Figure 1. Result of the questionnaire. The difference between Q5 and Q6 indicate a shift in the students' confidence to be able to apply AI concepts and techniques to impact the SDGs.

3.2 Assessing learners' sustainable competencies

3.2.1 Motivation and prior knowledge

Most students enrolled in the course perceive artificial intelligence (AI) as a beneficial tool for promoting a more sustainable future. Among the participants, 16 out of 30 expressed agreement or strong agreement with an interest in both dimensions of Sustainable AI. Conversely, only two participants

displayed a strong preference for one aspect over the other. While a substantial number of students feel competent in assessing their personal environmental impact (AB04), only half were able to accurately rank four activities by their environmental impact (GE08). Furthermore, thirteen students highlighted resource requirements, particularly energy consumption, as significant obstacles to achieving the Sustainable Development Goals (SDGs). Additionally, ten students identified bias as a problematic issue, and seven expressed concerns that AI could potentially exacerbate social disparities instead of alleviating them.

3.2.2 Willingness

The covariance matrix presented in Fig. 2 indicates that item WI02 shows poor correlation with the other items within the scale, suggesting that it should be replaced in future assessments. Additionally, a notable correlation between AB05 and WI03 with the behavior-related items (BE01-BE05) suggests that political and communal engagement are closely linked to behavioral outcomes. Summary statistics derived from four questions assessed learners' disposition towards engaging with sustainability. The calculated Cronbach's alpha coefficient of 0.75 demonstrates good internal consistency among the items used. Furthermore, a cumulative mean of 1.48 and a standard deviation of 0.56 reflect a robust willingness among participants to promote sustainability.

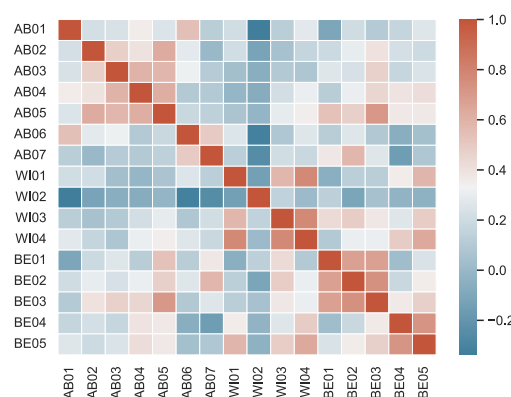


Figure 2. Willingness, abilities, and behavior: A correlation matrix of all Likert-scale items, organized according to their respective scales, illustrating the relationships among different dimensions.

3.2.3 Abilities

Six questions were utilized to assess learners' capabilities to engage with sustainability actions within the framework of Sustainable AI. The analysis yields a Cronbach's alpha coefficient of 0.76, indicating good internal consistency. Additionally, the responses have a cumulative mean of 0.59 and a standard deviation of 0.57, reflecting a moderate level of perceived ability to engagement with sustainability actions in the context of AI among the learners.

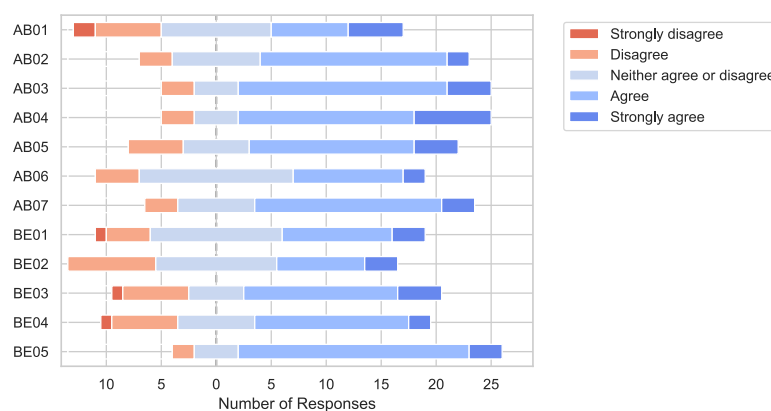


Figure 3. Abilities and behaviors: Among the surveyed abilities, the most notably deficient is the capacity to apply AI concepts effectively towards influencing Sustainable Development Goals (SDG) (AB01). Conversely, students report active efforts to minimize their ecological footprint (BE05).

3.2.4 Behaviors

The final set of five questions was employed to evaluate learners' behaviors concerning sustainability. This subset achieved a Cronbach's alpha coefficient of 0.79, indicating good internal consistency. The cumulative mean for this category was recorded at 0.43 with a standard deviation of 0.70, reflecting a moderate variability in behaviors related to sustainability among the learners. Fig. 3 displays the participants answers for the abilities and behaviors category.

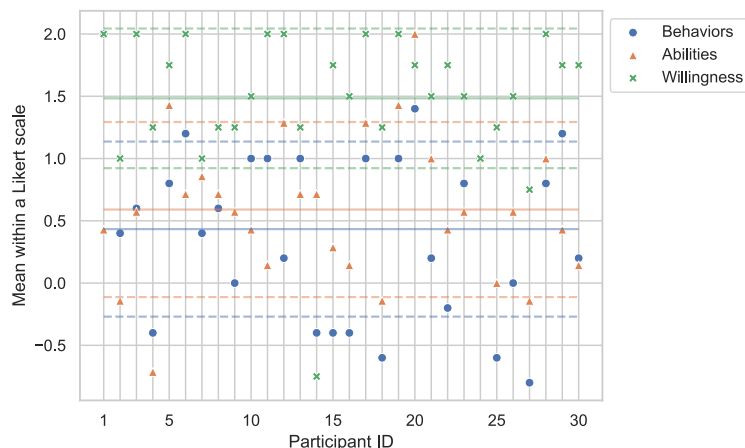


Figure 4. Mean of willingness, abilities, and behaviors for each participant. Solid lines indicate the mean of the participants' value (mean of the means) and dashed line the respective standard deviation. Willingness is above abilities, which is above behavior.

In Fig. 4, a comprehensive overview is presented. Each data point signifies the average score of an individual participant with respect to one of the three Likert scales. Notably, willingness appears robust among participants (mean=1.48, std=0.56) from the summer semester 2024, whereas their abilities regarding Sustainable AI are comparatively weaker (mean=0.59, std=0.57). This outcome aligns with expectations, as students typically enroll in a course focused on sustainability primarily due to a pre-existing inclination towards the subject. Moreover, it is unreasonable to presuppose existing competencies in Sustainable AI at the outset. Nonetheless, responses to items G06 and G07 indicate that students already possess conceptual ideas about leveraging AI technology and understanding its potential risks.

3.3 Adaptations

Using the feedback of our coaches, we implemented some changes for the current and next semesters to continually improve and adapt the course and address the feedback encountered in the last two semesters.

The timeline of the course was adopted to the universities' semester schedules, especially to avoid overlapping the workshop week with exam times. Therefore, students can concentrate during the workshop week and are not distracted to other obligations. To reduce the student workload during the workshop week, challenges for the current semester will be presented in the third week of classes. This allows students more time to focus on, research, and prepare for the issues and topics, facilitating more effective preparation for the intensive workshop week.

Given the broad scope of AI and sustainability, starting from summer semester 2024, each semester will focus on a specific theme within this area. This approach allows for a deeper and more intensive exploration of specific aspects of sustainability and AI.

4 CONCLUSIONS

The course yields mostly positive outcomes, as evidenced by the given feedback from students and partner organizations alike. Participants of the winter semester 2023/24 demonstrated high levels of engagement and commitment throughout the course, diligently working to meet the challenges presented. The comprehensive structure of the course, which included challenge-based learning mixed with expert inputs, feedback sessions, and support from challenge-giving organizations, although time-intensive, proved immensely beneficial for the students. This intensive interaction has not only fostered

a deeper understanding of the intersection between AI and sustainability but also enhanced the students' self-efficacy within this domain.

The course's interactive and interdisciplinary nature was particularly well-received. Students reported significant gains in their understanding of AI's potential to enhance sustainability efforts, as well as a sustained interest in exploring this field further. Additionally, the ability of participants to critically assess the challenges and benefits associated with the Sustainable Development Goals (SDGs) has notably increased, reinforcing the value of the course's challenge-based learning framework.

In terms of motivation and abilities of the current cohort, although many students feel confident in assessing their personal environmental impact, there remains a gap in accurately understanding and prioritizing activities based on their environmental footprint. The findings also suggest a hierarchy in the learning outcomes, where willingness to engage with sustainable practices outstripped actual abilities, which in turn were higher than demonstrated behaviors. This finding is consistent with the expectations for a course focused on sustainability, attracting students already inclined towards environmental issues.

In conclusion, the course has effectively cultivated a cohort of informed and motivated students equipped with critical skills to address challenges in the dynamic fields of sustainability and AI. The questionnaires developed for this program provide a valuable framework for educators to assess and profile sustainability competencies, though further refinement and ongoing evaluation are recommended to capture comprehensive learning trajectories and outcomes.

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